

SCORE CENTERING STABILIZES
OFF-POLICY REINFORCEMENT LEARNING

[\[2609.20807\] Score Centering Stabilizes
Off-policy Reinforcement Learning](#)

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Score Centering Stabilizes Off-policy RL

东川路第一可爱猫猫虫

主要内容

- 训推不一致及其来源
- 重要性采样可以解决但引入高方差噪声
- 训推不一致时策略梯度更新的分解
- Score Centering方法

策略梯度

- 对LLM训RL有三步骤：

采样rollout 给每条答案以奖励 计算梯度

- $\nabla_{\theta} \mathbb{E}_{p_{\theta}}[R] = \mathbb{E}_{p_{\theta}}[R \nabla_{\theta} \log p_{\theta}(y)]$

y是当前模型采样的rollout R是y获得的奖励

$\nabla_{\theta} \log p_{\theta}(y)$ 在论文里被称为score

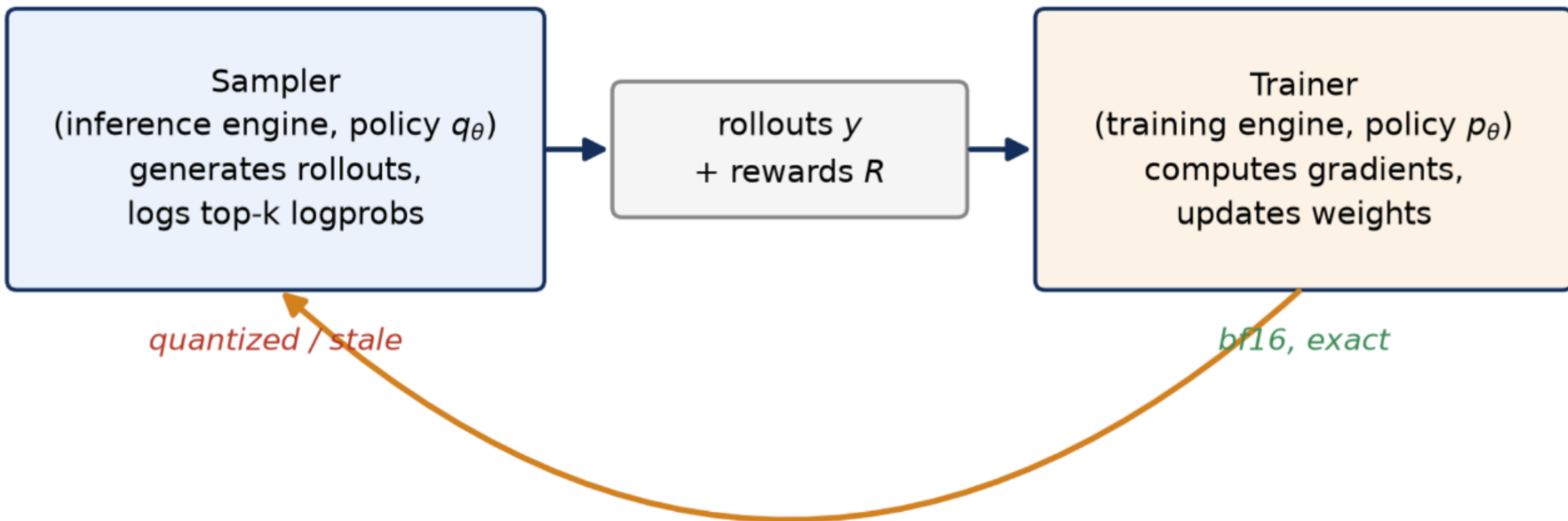
期望奖励的梯度=score以奖励R加权后求期望

- 等式右边需要做两次独立的反向传播

模型作为sampler来采样rollout

模型作为trainer来计算梯度

Same weights θ , two programs: $q_\theta \neq p_\theta$ (training-inference mismatch, TIM)



训推不一致及其来源

- 同一个LLM先后作为sampler p和trainer q
二者的输出理应完全一致，但实际中几乎从不如此
- 训推引擎之间存在差异

training-inference mismatch (TIM)

来源

精度不同或代码库不同

浮点运算的非结合律 fp4 $0.5+0.5+2$

staleness 异步RL 训推解耦

重要性采样能解决TIM, but at what cost?

- $\mathbb{E}_{z \sim \pi_p} [f(z)] = \mathbb{E}_{z \sim \pi_q} \left[\frac{\pi_p(z)}{\pi_q(z)} f(z) \right]$

$$\mathbb{E}_{z \sim \pi_q} \left[\frac{\pi_p(z)}{\pi_q(z)} f(z) \right]$$

$$\text{GRPO里} \frac{\pi_{\theta}(y_{i,t} | x, y_{i,<t})}{\pi_{\theta_{\text{old}}}(y_{i,t} | x, y_{i,<t})}$$

应该多次采样, 但GRPO只能采样一次

- 在 $x, y_{i,<t}$ 的条件下, 生成next token的过程只会发生一次
只采样了 $y_{i,t}$, 相当于把单次采样的结果当成了期望
因此引入了高方差噪声

On policy且奖励恒定时 理论上不学习

- 在一个固定前缀 $y_{<t}$ 上, 采样器 q 产生token y_t , v 遍历词表
 p_v, q_v 分别是trainer和sampler对 v 的下一token概率
 $s_v = \nabla_{\theta} \log p_v$ 是它的 score
 $\bar{s} = \sum_v q_v s_v$ 是采样器 q 下的期望 score
- $\mathbb{E}_p[s_{y_t}] = \sum_v p_v \nabla_{\theta} \log p_v = \sum_v \nabla_{\theta} p_v = \nabla_{\theta} \sum_v p_v = \nabla_{\theta} 1 = 0$
score函数期望为0
前提是on policy, 采样和求梯度用同一个p
- 如果在奖励恒定的环境里做原始策略梯度的RL
 $\mathbb{E}_p[R s_{y_t}] = R \mathbb{E}_p[s_{y_t}] = R \cdot 0$

TIM下的期望更新可以分解为信号和漂移

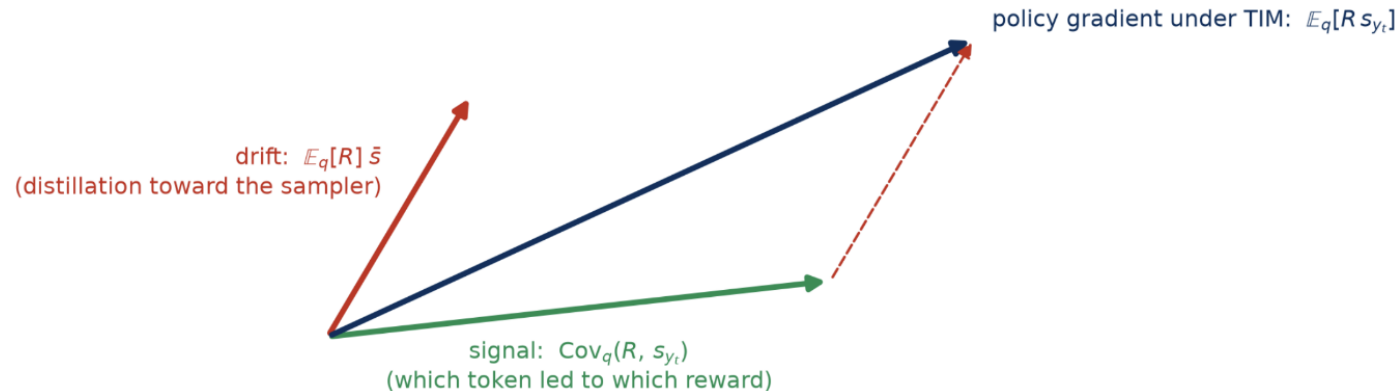
- 在TIM情况下，期望梯度不为0
- On policy情况下

$$\mathbb{E}_p[s_{y_t}] = \sum_v p_v \nabla_{\theta} \log p_v = \sum_v \nabla_{\theta} p_v = \nabla_{\theta} \sum_v p_v = \nabla_{\theta} 1 = 0$$

$$\text{TIM情况下 } \bar{s} = \mathbb{E}_q[s_{y_t}] \neq 0$$

- $$\underbrace{\mathbb{E}_q[R s_{y_t}]}_{\text{policy gradient update}} = \underbrace{\mathbb{E}_q[R] \bar{s}}_{\text{"drift"}} + \underbrace{\text{Cov}_q(R, s_{y_t})}_{\text{"signal"}}$$

policy gradient under TIM: $\mathbb{E}_q[R s_{y_t}]$

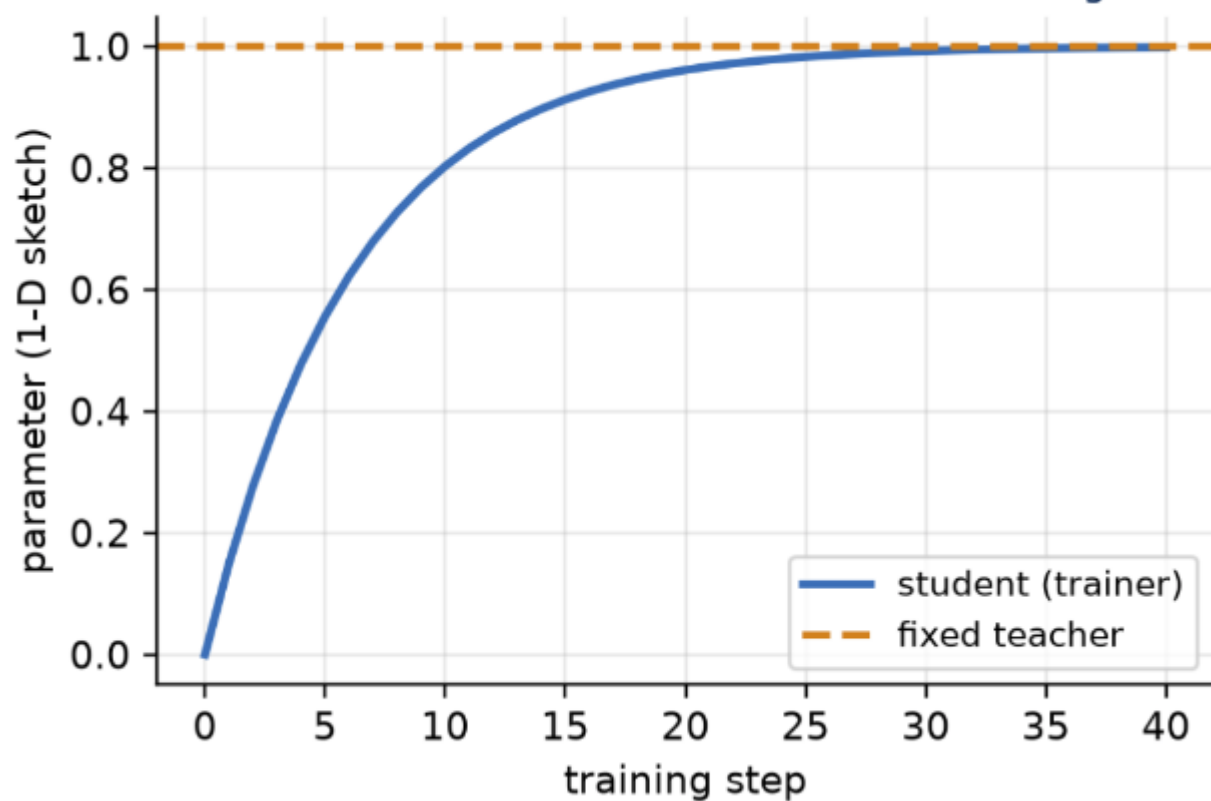


Drift is an artifact of TIM that acts as distillation toward the sampler. The drift term carries no information about which rollouts were successful: it depends on the rewards only through their mean $\mathbb{E}_q[R]$. Only the covariance term sees which token led to which reward. What the drift term does instead is determined by its direction $\bar{s}$: the expected score is the negative gradient of the cross-entropy (SFT) loss with the sampler as the teacher, so vanilla policy gradient distills the trainer toward the sampler at every prefix, scaled by the expected reward at that prefix. This is purely an

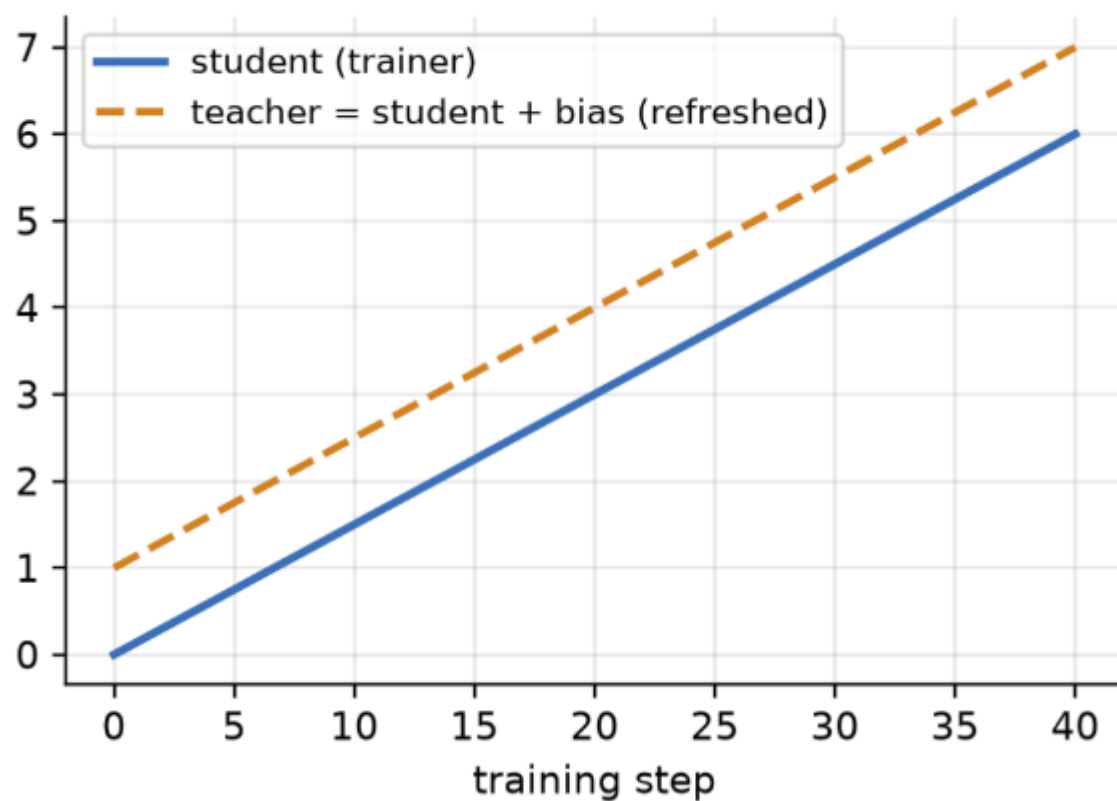
- 以 q 为目标分布的交叉熵损失 $CE(q, p) = -\sum_v q_v \log p_v$
 它对 θ 的梯度是 $-\sum_v q_v \nabla_{\theta} \log p_v = -\bar{s}$
 沿 $+\bar{s}$ 走一步 = 沿交叉熵的负梯度走一步 = 让 p 更像 q
漂移是TIM的副产物，其作用相当于向sampler q 蒸馏
- $$\underbrace{\mathbb{E}_q[R s_{y_t}]}_{\text{policy gradient update}} = \underbrace{\mathbb{E}_q[R] \bar{s}}_{\text{``drift''}} + \underbrace{\text{Cov}_q(R, s_{y_t})}_{\text{``signal''}}$$

 on policy时, drift=0, p 向 q 蒸馏, 这是无害的
 off policy时, q 是 p 的带偏差的拷贝
 每一步都把 p 推向 q , p 的权重又被同步回 q
 误差会循环累加

Offline distillation: fixed teacher -> converges



Online RL under TIM: teacher tracks student -> drifts forever



Score Centering

Score centering removes drift. These observations directly motivate our method, called *score centering*: even under TIM, we want the expected score to be zero at every prefix, as it is on-policy. We achieve this simply by subtracting from each score the expected score under the sampler, replacing s_{y_t} by the centered score $\tilde{s}_{y_t}$:

$$\bullet \tilde{s}_{y_t} = s_{y_t} - \bar{s}, \quad \bar{s} = \sum_v q_v s_v = \sum_v q_v \nabla_{\theta} \log p_v$$

$$\begin{aligned} \mathbb{E}_q[R \tilde{s}_{y_t}] &= \overset{\mathbb{E}_q[\tilde{s}_{y_t}] = \bar{s} - \bar{s} = 0}{\cancel{\mathbb{E}_q[R] \mathbb{E}_q[\tilde{s}_{y_t}]}} + \text{Cov}_q(R, \tilde{s}_{y_t}) \\ &= \text{Cov}_q(R, \tilde{s}_{y_t}) \\ &= \text{Cov}_q(R, s_{y_t}) \quad \longleftarrow \begin{array}{l} \text{only difference to on-policy update} \\ \text{is the covariance sampling distribution} \end{array} \end{aligned}$$